

# Constant-coefficient abelian 1518-extensions of the $\mathbb{Z}/3$ shift are direct products

hence none refutes  $1518 \Rightarrow 47, 614, 817, 3862$ , and the ETP’s 15-element countermodel  
is an extension of the same base with base-dependent coefficients

Carlo Perassi\*

4 September 2026 — draft for discussion

## 1 Setting

Laws are numbered as in the Equational Theories Project (ETP), `teorth/equational_theories` at commit `88088faa`. Law 1518 is

$$x = (y \diamond y) \diamond (x \diamond (y \diamond x)),$$

and the four one-variable targets are

$$\begin{array}{ll} \text{E47 : } x = x \diamond (x \diamond (x \diamond x)), & \text{E614 : } x = x \diamond (x \diamond ((x \diamond x) \diamond x)), \\ \text{E817 : } x = x \diamond ((x \diamond x) \diamond (x \diamond x)), & \text{E3862 : } x \diamond x = (x \diamond (x \diamond x)) \diamond x. \end{array}$$

The smallest finite countermodel to  $1518 \Rightarrow 47$  has 15 elements (Brox and Le Floch, Lean Zulip, October 2025), and the ETP’s 15-element table `Refutation939` refutes all four at once. Every 1518-magma with at most 7 elements satisfies all four (checked here on the complete census of Section 8).

Following the blueprint chapter “Magma cohomology”, an *extension* of a magma  $G$  by an abelian group  $M$  with  $\alpha, \beta \in \text{End}(M)$  and  $f : G \times G \rightarrow M$  is the magma on  $G \times M$  with

$$(x, s) \diamond (y, t) = (x \diamond y, \alpha s + \beta t + f(x, y)). \quad (1)$$

The coefficients  $\alpha, \beta$  are constant. If  $G$  and the linear magma  $(M, \alpha s + \beta t)$  both satisfy a law  $E$ , the extension satisfies  $E$  if and only if  $f$  is an *E-cocycle*, a linear condition on  $f$ . Refuting  $E \Rightarrow E'$  by this method means exhibiting an *E-cocycle* that is not an *E'-cocycle*.

In November 2024 this method produced the 12-element model for  $879 \not\Rightarrow 4065$  (Tao, Lean Zulip, *Austin pairs*, message 484839156: base  $(b, i) \diamond (c, j) = (b + c, j + 1)$  on  $\mathbb{Z}/2 \times \mathbb{Z}/3$ , fibre  $\mathbb{Z}/2$ ,  $\alpha = \beta = 1$ ) and failed for 1518 (Bolan 485027669; Tao 485133886: “perhaps sometimes the linear algebra just doesn’t work out”). Tao also observed (485146097) that since the targets have one variable, the base may be taken one-generated, and conjectured (485148288) that the only nontrivial one-generated 1518-magma satisfying the four targets is the cyclic shift  $x \diamond y = y + 1$  on  $\mathbb{Z}/3$ .

---

\*Computation and drafting with AI assistance under the author’s direction. Everything below is reproducible from a small standard-library script set; see Section 11.

## 2 Statement

**Theorem 1.** *Let  $G$  be the three-element magma  $x \diamond y = y + 1$  on  $\mathbb{Z}/3$ , or the one-element magma. Let  $M$  be a finite abelian group and  $\alpha, \beta \in \text{End}(M)$  such that  $(M, \alpha s + \beta t)$  satisfies 1518. Then every 1518-cocycle  $f: G \times G \rightarrow M$  is a 47-cocycle, a 614-cocycle, an 817-cocycle and a 3862-cocycle. Consequently no extension (1) of  $G$  refutes  $1518 \Rightarrow E'$  for  $E' \in \{47, 614, 817, 3862\}$ .*

**Corollary 2.** *Let  $H$  be a finite 1518-magma and suppose an extension (1) of  $H$  by a finite  $M$  refutes  $1518 \Rightarrow E'$  for one of the four targets. Then  $H$  contains a one-generated submagma with at least 9 elements. Dually, the same holds for  $2054 \Rightarrow 255, 2847, 2644, 3456$  with the shift  $x \diamond y = x + 1$ .*

The corollary combines the theorem with Tao’s reduction and with a census (Section 8): up to isomorphism the one-generated 1518-magmas with at most 8 elements are the trivial magma and the  $\mathbb{Z}/3$  shift. Section 6 removes the size bound: one-generated magmas satisfying 1518 and 3862 are trivial or the shift in any size, and the obstruction becomes unconditional for finite bases (Theorem 5).

## 3 Proof of Theorem 1

**Step 1: the finite-fibre ring.** Expanding 1518 on the linear magma  $s \diamond t = \alpha s + \beta t$  and comparing the coefficients of  $x$  and  $y$  gives the identities in  $\text{End}(M)$

$$\beta\alpha + \beta^3 = 1, \quad \alpha^2 + \alpha\beta + \beta^2\alpha = 0. \quad (2)$$

The first reads  $\beta(\alpha + \beta^2) = 1$ , so  $\beta$  is surjective;  $M$  is finite, hence  $\beta$  is bijective and  $\alpha + \beta^2 = \beta^{-1}$ , i.e.  $\alpha = \beta^{-1} - \beta^2$ . In particular  $\alpha$  commutes with  $\beta$ . Substituting into the second identity and multiplying by  $\beta^2$  yields

$$\beta^5 + \beta^3 - \beta^2 - 1 = (\beta^2 + 1)(\beta^3 - 1) = 0.$$

Conversely, these two facts imply (2). Finiteness of  $M$  is used only here, to pass from “surjective” to “bijective”; everything below holds verbatim for any abelian group  $M$  on which  $\beta$  is injective. Therefore  $M$  is a module over the commutative ring

$$R = \mathbb{Z}[b]/(b^5 + b^3 - b^2 - 1), \quad a := b^4 - b = b^{-1} - b^2, \quad b^{-1} = b^4 + b^2 - b, \quad (3)$$

via  $b \mapsto \beta$ , and then  $a \mapsto \alpha$ .  $R$  is free of rank 5 over  $\mathbb{Z}$ , and every  $R$ -module is an admissible fibre. Over an integral domain this recovers Nielsen’s classification (485154828):  $\alpha = 0$ ,  $\beta^3 = 1$ , or  $\beta = 1 - \alpha$ ,  $\alpha^2 - 2\alpha + 2 = 0$ .

**Step 2: cocycle conditions over  $R$ .** For any word  $w$ , the fibre component of  $w$  in the extension is  $\sum_i P_{w,i}(a, b) s_i + \sum Q(a, b) f(\cdot, \cdot)$ , the sum running over the non-variable subterms (blueprint). The linear magma over  $R$  satisfies 1518 and each of the four targets (the coefficient polynomials  $P$  of both sides agree in  $R$ ; checked), so the  $s_i$  terms cancel and the  $E$ -cocycle condition is a matrix  $D_E$  over  $R$  acting on the nine unknowns  $f_{xy} = f(x, y)$ . For every  $R$ -module  $M$ ,  $Z_E^2(G, M) = \ker(D_E: M^9 \rightarrow M^k)$ .

Writing indices modulo 3, the 1518 condition at the base pair  $(x, y)$  is

$$(b^4 - b) f_{yy} + b^2 f_{yx} + b f_{x, x+1} + f_{y+1, x+2} = 0, \quad (4)$$

nine equations (when  $x = y$  the first two terms merge into  $(b^4 - b + b^2)f_{xx}$ ). The target conditions at  $x$  are

$$\text{E47: } b^2 f_{xx} + b f_{x,x+1} + f_{x,x+2} = 0, \quad (5)$$

$$\text{E614: } (b - b^4)f_{xx} + b f_{x,x+1} + f_{x,x+2} + b^2 f_{x+1,x} = 0, \quad (6)$$

$$\text{E817: } (1 + b^2 - b^3)f_{xx} + f_{x,x+2} + b f_{x+1,x+1} = 0, \quad (7)$$

$$\text{E3862: } b^3 f_{xx} + (b - b^4)f_{x,x+1} - f_{x+2,x} = 0. \quad (8)$$

**Step 3: containment by certificate.** For each target  $E'$  there is a  $3 \times 9$  matrix  $C_{E'}$  over  $R$  with  $C_{E'} D_{1518} = D_{E'}$  entrywise in  $R$ . Then  $D_{1518}f = 0$  implies  $D_{E'}f = C_{E'} D_{1518}f = 0$  for every  $R$ -module  $M$ , which is the theorem. Writing  $r_{xy}$  for the 1518 row (4) at  $(x, y)$ , the 47 row (5) at  $x = 0$  is

$$D_{47}(0) = r_{00} + (b - b^2 - b^4)r_{11} + (-1 + b + b^3)r_{20}. \quad (9)$$

The full certificates have 15 to 18 nonzero entries per target; they are listed in the accompanying file `12_certificate_3.json`, and the one-element base has  $1 \times 1$  certificates in `12_certificate_1.json`.

*Remark.* The certificates were found, not guessed. Over the rank-5 ring the  $R$ -span of the nine 1518 rows is a  $\mathbb{Z}$ -lattice of rank 35 in  $\mathbb{Z}^{45}$ , and membership of the target rows was decided by integer row reduction. Containment for all  $R$ -modules is in fact *equivalent* to this lattice containment (take  $M = R^9 / \text{rowspan } D_{1518}$  and a finite quotient), so a failure would have produced an explicit finite countermodel. As a by-product,  $Z_{1518}^2(G, M)$  has  $R$ -rank 2, and all of it consists of target cocycles.

## 4 The stronger statement: $H^2$ vanishes

Theorem 1 is a corollary of a cleaner fact. A *coboundary* is a function of the form  $f(x, y) = g(x \diamond y) - \alpha g(x) - \beta g(y)$  for some  $g: G \rightarrow M$ ; coboundaries are cocycles for every law, and an extension by a coboundary is isomorphic to the direct product via  $(x, s) \mapsto (x, s + g(x))$  (blueprint, cohomology chapter). Let  $E: M^3 \rightarrow M^9$  be the coboundary map  $g \mapsto (g(y+1) - \alpha g(x) - \beta g(y))_{x,y}$ , a  $9 \times 3$  matrix over  $R$ .

**Theorem 3.** *For the  $\mathbb{Z}/3$  shift  $G$  and every finite abelian group  $M$  with  $\alpha, \beta$  as in Theorem 1, every 1518-cocycle  $G \times G \rightarrow M$  is a coboundary:  $H_{1518}^2(G, M) = 0$ . Consequently every extension (1) of  $G$  by such a fibre that satisfies 1518 is isomorphic to the direct product  $G \times (M, \alpha s + \beta t)$ , and satisfies every law that both factors satisfy.*

*Proof.* There are matrices  $P$  ( $3 \times 9$ , two nonzero entries) and  $Q$  ( $9 \times 9$ ) over  $R$  with

$$EP + QD_{1518} = I_9 \quad \text{entrywise in } R. \quad (10)$$

If  $D_{1518}f = 0$  then  $f = EPf + QD_{1518}f = E(Pf)$ , a coboundary, for every  $R$ -module  $M$ . The matrices are in `h2_certificate.json`; identity (10) is re-verified by multiplication in  $R$  and kernel-checked in `H2Cert.lean`.  $\square$

Theorem 1 follows: the shift and every finite linear 1518-magma satisfy 47, 614, 817 and 3862, so their direct products do, so every extension does. The same argument gives the conclusion for *any* law satisfied by the shift and by all finite linear 1518-magmas.

Three checks and one contrast. At every root  $r$  of  $b^5 + b^3 - b^2 - 1$  in  $\mathbb{F}_p$ ,  $p \leq 47$ , one finds  $\dim Z_{1518}^2 = \dim B^2 = 2$ . Over the non-field fibre  $\mathbb{Z}/4$  with  $\beta = 1$ , brute force gives 16 cocycles

and 16 coboundaries. The one-element base behaves differently: there  $H^2$  can be nonzero (for  $M = \mathbb{Z}/3$ ,  $\beta = 1$ ,  $\alpha = 0$  every constant is a cocycle and no nonzero constant is a coboundary), yet the containment of Theorem 1 still holds by its own certificate. Finally, for 879 over the affine base  $3x + y + 1 \bmod 6$  with fibre  $\mathbb{Z}/2$ ,  $\alpha = \beta = 1$ , one finds  $\dim Z^2 = 7$ ,  $\dim B^2 = 5$ , so  $\dim H^2 = 2$ : that is the room the method used to produce the 12-element model, and it is absent for 1518 over the shift.

## 5 Consequences, and where the obstruction ends

**The exact set of unreachable non-implications.** By Theorem 3, a constant-coefficient 1518-extension of the shift satisfies every law  $E'$  that holds in the shift and in every finite linear 1518-magma; the second condition is decided once and for all in  $R$  (the coefficient polynomials of  $E'$  agree in  $R$ ). Reading the row of 1518 in the ETP’s closed finite implication graph (`finite_graph.json`, fetched 2026-09-04): 1518 has 5 finite implications and 4689 finite non-implications, none unknown. Of the non-implications, 4347 are refuted by the shift itself, 329 by some finite linear 1518-magma on its own, and exactly

$$13 : \quad 47, 65, 375, 614, 632, 679, 817, 872, 879, 1491, 3862, 3915, 4118$$

are unreachable by every constant-coefficient extension of the shift: four one-variable laws (the targets) and nine two-variable laws. As a control, all 5 finite implications pass both checks. For the nine two-variable laws the statement is about this base only; the reduction to one-generated bases behind Corollary 2 needs a one-variable target.

**The census, twice.** Corollary 2 rests on the claim that the only one-generated 1518-magmas with at most 8 elements are the trivial one and the shift. Two independent tools agree: Mace4 (complete labelled enumeration at sizes 2–4, complete up to isomorphism at 5–7) and z3 with a direct reachability encoding of “generated by one element” (satisfiable only at size 3, where it returns the shift; unsatisfiable at sizes 2 and 4 to 8). The two tools also agree on the labelled counts 1, 6, 54, 1500 at sizes 2 to 5. Sizes 2 to 8 are moreover Lean theorems: the same encoding, discharged by `bv_decide`, whose trust rests on Lean’s verified LRAT checker run natively (Section 8).

**The ETP witness is an extension of the shift with base-dependent coefficients.** The 15-element table `Refutation939` has a unique nontrivial congruence, with three classes of five elements, and its quotient is isomorphic to the  $\mathbb{Z}/3$  shift. Labelling each class by  $\mathbb{Z}/5$  (the labelling is unique up to an affine relabelling of each fibre) writes the operation as

$$(x, s) \diamond (y, t) = (y + 1, \alpha_{x,y} s + \beta_{x,y} t + c_{x,y}),$$

with, for rows  $x = 0, 1, 2$  and columns  $y = 0, 1, 2$ ,

$$(\alpha_{x,y}, \beta_{x,y}, c_{x,y}) = \begin{pmatrix} (2, 4, 0) & (2, 3, 2) & (4, 1, 2) \\ (4, 2, 0) & (2, 4, 0) & (1, 4, 2) \\ (1, 3, 4) & (4, 2, 0) & (1, 2, 1) \end{pmatrix}.$$

The coefficients take the values  $\alpha_{x,y} \in \{1, 2, 4\}$ ,  $\beta_{x,y} \in \{1, 2, 3, 4\}$  and change with the base pair. This is exactly the general form of the blueprint’s Lemma “No counterexamples via linear extension” for 677, with the shift as base. By Theorem 3 no relabelling can make the coefficients constant. So the line is sharp: over the  $\mathbb{Z}/3$  shift, constant coefficients give only direct products, while base-dependent coefficients give the ETP’s minimal countermodel. For 677 the obstruction survives base-dependent coefficients (Tao, June 2025); for 1518 over the shift it does not.

## 6 The one-generated bases are classified

The census left open what happens above size 8. It does not stay open. Write  $S(x) = x \diamond x$ .

**Theorem 4.** *In every magma satisfying 1518 and 3862, for every  $x$  and all  $u, v$  in the set  $\{x, S(x), S(S(x))\}$  one has  $u \diamond v = S(v)$ , and  $S(S(S(x))) = x$ . Consequently the set is closed under  $\diamond$ , carries the multiplication table of the cyclic shift, has one element or three, and every one-generated magma satisfying 1518 and 3862 is the trivial magma or the  $\mathbb{Z}/3$  shift.*

Vampire 5.1.0 found equational proofs of these nine equalities in milliseconds from 1518 and 3862 alone, by superposition and demodulation (9 to 32 proof lines each, `atp/eq*.proof`); they were transcribed into core Lean 4.33.1 (`lean/OneGenerated1518.lean`): the table and the closure of words in  $x$  are proved with no axioms at all, and the one-or-three-elements statement with `propext`, `Classical.choice`, `Quot.sound` only. The classification therefore needs neither finiteness nor the SAT census. This proves Tao’s conjecture of 2024-11-29, which assumed all four targets; assuming 3862 alone is equivalent, since 3862 implies the other three (Bolan, 485153888, the same day). The target cannot be dropped: `Refutation939` is a one-generated 1518-magma with 15 elements, and it violates 3862 (60 of its 135 closure products leave the set). As a control, on all 41 319 six-element 1518-magmas the largest one-generated submagma has 3 elements.

Vampire also shows that under 1518 together with left injectivity and left surjectivity, which hold in every finite 1518-magma, the four targets 47, 614, 817, 3862 are pairwise equivalent (twelve theorems), and that equationally 3862 implies the other three. This re-derives observations from the *Austin pairs* thread: Tao showed that in a finite 1518-magma the squaring map and all left multiplications are bijective and that 47 and 614 are equivalent via 359 (484117783), Bolan that 3862 implies the other three (485153888), and Nielsen, building on Tencer (484863266), that under left cancellation the four targets are equivalent to each other and to  $S^3x = x$  (485248752); Le Floch restated the finite case in October 2025. Hence:

**Theorem 5** (full obstruction, finite bases). *Let  $H$  be a finite 1518-magma,  $M$  a finite abelian group with  $\alpha, \beta$  making  $\alpha s + \beta t$  a 1518-magma, and  $f$  a 1518-cocycle on  $H$ . If the extension  $H \times_f M$  violates one of 47, 614, 817, 3862, then  $H$  already violates it.*

*Proof.* If  $H$  satisfies the target then, being finite, it satisfies 3862; by Theorem 4 every one-generated submagma  $\langle x \rangle$  is trivial or the shift; a violation at  $(x, s)$  would live in the sub-extension over  $\langle x \rangle$ , which satisfies the four targets by Theorems 1 and 3.  $\square$

So constant-coefficient abelian cohomology never refutes any of these four implications from any finite base. Non-vanishing  $H^2$  over other bases is real:  $H^2_{1518}$  is nonzero for the right projection on two elements and for every 1518-base of sizes 4 and 5, and such extensions do refute two-variable laws ( $1518 \Rightarrow 632$  and  $1518 \Rightarrow 879$  at carrier 6, from the base  $[[0, 1, 2], [0, 1, 2], [1, 0, 2]]$  with fibre  $\mathbb{Z}/2$ ), but by Theorem 5 they cannot touch the one-variable targets. The remaining prose in this section is the finite-base equivalence of the four targets and the assembly of Theorem 5.

## 7 Base-dependent extensions of the shift: an explicit family

Question 3 below asked whether the 15-element models are the generic case of something. For two of the six, over the shift, the answer is yes. Let  $(x, s) \diamond (y, t) = (y + 1, a_{xy}s + b_{xy}t + c_{xy})$  on  $\mathbb{Z}/3 \times \mathbb{F}_p$  with coefficients depending on the base pair. Law 1518 is equivalent to two families of identities in  $(a, b)$ ,  $b_{y+1, x+2} (a_{x, x+1} + b_{x, x+1} b_{y, x}) = 1$  and  $a_{y+1, x+2} d_y + b_{y+1, x+2} b_{x, x+1} a_{y, x} = 0$  with  $d_z = a_{zz} + b_{zz}$ ,

together with a homogeneous linear system for  $c$ ; whether the four targets hold depends only on  $(a, b)$ . The second family links the entries of  $a$  along the three orbits of  $(u, v) \mapsto (u + 2, v + 1)$  on  $\mathbb{Z}/3 \times \mathbb{Z}/3$ , so  $a$  is determined by  $b$ , and the whole solution set can be enumerated in  $O(p^5)$  steps (`basedep_direct.py`).

Rescaling the fibre over each base point,  $s \mapsto \lambda_x s$ , is an isomorphism of extensions; the group  $(\mathbb{F}_p^*)^3/\mathbb{F}_p^*$  acts freely on the solutions. The number of solutions is  $8(p-1)^2$  when  $p \equiv 1 \pmod{4}$  and  $4(p-1)^2$  when  $p \equiv 3 \pmod{4}$ , of which  $2(p-1)^2$ , respectively none, refute the targets (all  $p \leq 61$ ; the formulas were recorded before  $p = 37, 41, 43, 47, 53, 61$  were run). So there are eight, respectively four, orbits. Lifting the orbit representatives (normalised by  $b_{00} = b_{11} = 1$ ) to  $\mathbb{Q}(i)$  by the Chinese remainder theorem over  $p = 13, 17, 29, 37, 41$  and rational reconstruction gives eight tables in which 1518 holds exactly in  $\mathbb{Q}(i)$  (`basedep_family.py`): the direct product ( $a = 0, b = 1$ ); three rational tables forming one orbit under the base rotation, with entries in  $\mathbb{Z}[1/3]$ ; a Galois-conjugate pair with  $D = 1$  and entries in  $\mathbb{Z}[i]$ ; and a Galois-conjugate pair with  $D = \pm i$ , where  $D = d_0 d_1 d_2$  is the scalar by which  $S^3$  acts on the fibre. The six tables with  $D = 1$  satisfy all four targets exactly; the last two refute them. In particular the construction gives nothing for  $p \equiv 3 \pmod{4}$  (proved for  $p \leq 61$  by exhaustion, and over  $\overline{\mathbb{Q}}$  by a Gröbner basis: after normalising  $b_{00} = b_{11} = 1$  and inverting all  $b_{xy}$  and  $d_z$ , the ideal of the identities is zero-dimensional of degree 8, so no further solutions exist in characteristic 0, nor in all but finitely many positive characteristics; the same computation over  $\mathbb{F}_p$  gives degree 8 for every prime  $5 \leq p \leq 257$ ).

**Theorem 6** (explicit family). *Let  $R$  be a commutative ring in which 2 is invertible and  $i \in R$  satisfies  $i^2 = -1$ , and let  $M \neq 0$  be an  $R$ -module. On  $\mathbb{Z}/3 \times M$  define  $(x, s) \diamond (y, t) = (y + 1, a_{xy}s + b_{xy}t)$  with*

$$a = \begin{pmatrix} -\frac{1+i}{2} & -\frac{1}{2} & \frac{1-i}{2} \\ \frac{1-i}{2} & -\frac{1+i}{2} & 1+i \\ 1+i & \frac{1-i}{2} & 2i \end{pmatrix}, \quad b = \begin{pmatrix} 1 & \frac{1+i}{2} & -(1+i) \\ \frac{1}{2} & 1 & -2i \\ \frac{1+i}{2} & \frac{1}{2} & -2(1+i) \end{pmatrix}.$$

*Then 1518 holds, and each of 47, 614, 817, 3862 fails.*

*Proof.* The two families of identities hold exactly in  $\mathbb{Q}(i)$ , hence in  $\mathbb{Z}[i, 1/2]$  and in  $R$ . Each target law, evaluated on  $\mathbb{Z}/3 \times M$  with symbolic fibre coordinates, fails by a coefficient difference that is a unit of  $\mathbb{Z}[i, 1/2]$ :  $(1+i)/2$  for 47 and 614, 1 for 817,  $-2+2i$  and  $-i$  for 3862. A unit kills no nonzero element of  $M$ .  $\square$

The sizes are  $3|M|$ : 15 for  $M = \mathbb{F}_5$  (models 5 and 6 of the six enumerated by Brox, isomorphism checked), 27 for  $\mathbb{F}_9 = \mathbb{F}_3[i]$ , 39, 51, 75 ( $\mathbb{Z}/25$ ), 87, 111, 123, 147 ( $\mathbb{F}_{49}$ ), and so on; the 27- and 75-element members were re-checked by brute force. Since  $d = (\frac{1-i}{2}, \frac{1-i}{2}, -2)$ ,  $S^3(x, s) = (x, is)$  and the squaring map has order exactly 12. In the five rotation-invariant classes, and only there,  $S$  is an endomorphism, the constraint Le Floch used to find the 15-element model. Kernel-`decide` Lean files (`lean/FamilyF5.lean`, `lean/FamilyF13.lean`, no axioms) check 1518, the failure of the four targets,  $S^3 \neq \text{id}$  and  $S^{12} = \text{id}$  for the 15- and 39-element members. In a finite 1518-magma  $S$  and all left multiplications are bijective and the single-variable targets are equivalent to  $S^3 x = x$  (Tao, Tencer, Nielsen, November 2024); Mace4 samples at sizes 4 to 9 show orders 1 and 3 only, as they must below size 15. Whether a finite 1518-magma can have squaring of order other than 1, 3, 12 is open.

## 8 Verification status

| Layer                                                                                    | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Certificate identities $C D_{1518} = D_{E'}$ in $R$                                      | Lean 4.33.1 kernel, <b>decide</b> on integer coefficient lists, file <code>L2Cert.lean</code> ; axioms: <b>propext</b> only                                                                                                                                                                                                                                                                                                                                                                                                                |
| Homotopy identity $EP + QD_{1518} = I_9$ in $R$ (Theorem 3)<br>$H^2 = 0$ numerically     | Lean 4.33.1 kernel, file <code>H2Cert.lean</code> ; axioms: <b>propext</b> only; also re-multiplied in Python<br>$\dim Z^2 = \dim B^2 = 2$ at all 39 prime specialisations;<br>$\mathbb{Z}/4$ , $\beta = 1$ : 16 cocycles, 16 coboundaries                                                                                                                                                                                                                                                                                                 |
| Same identities<br>Rows (4)–(5) are the cocycle conditions; $R$ is the finite-fibre ring | re-multiplication in $R$ in Python<br>written argument above; not formalized. Independently re-derived by a second program with noncommutative coefficients (words in $\alpha, \beta$ ), specialised to $R$ only at the end: the identities (2) and all rows agree exactly                                                                                                                                                                                                                                                                 |
| One-generated classification (Theorem 4)                                                 | core Lean 4.33.1, <code>lean/OneGenerated1518.lean</code> : <b>table</b> , <b>words_in.T</b> with no axioms; <b>one_or_three</b> with the three standard axioms; transcribed from Vampire’s equational proofs                                                                                                                                                                                                                                                                                                                              |
| Theorem 6                                                                                | exact arithmetic in $\mathbb{Q}(i)$ (Python <b>fractions</b> ); classification exhaustive for $p \leq 61$ ; members over $\mathbb{F}_5$ , $\mathbb{F}_{13}$ by Lean kernel <b>decide</b> (no axioms), over $\mathbb{F}_9$ , $\mathbb{Z}/25$ by brute force                                                                                                                                                                                                                                                                                 |
| Target equivalences under 1518, finite bases                                             | Vampire 5.1.0: twelve theorems with left-quasigroup axioms; equationally $3862 \Rightarrow 47,614,817$ ; not in Lean                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Unreachable set of Section 5                                                             | ETP <code>finite_graph.json</code> (sha256 <code>d609274e...</code> ), row 1518 decoded; control: the 5 finite implications pass the shift and $R$ checks                                                                                                                                                                                                                                                                                                                                                                                  |
| Structure of <code>Refutation939</code>                                                  | exhaustive search over labellings of its three classes by $\mathbb{Z}/5$ ; $20^3$ labellings give base-dependent affine form, none constant (Theorem 3 forbids it)                                                                                                                                                                                                                                                                                                                                                                         |
| Numeric cross-check                                                                      | at every root $r$ of $b^5 + b^3 - b^2 - 1$ in $\mathbb{F}_p$ , $p \leq 47$ (39 cases), fibre $\mathbb{Z}/p$ , $\beta = r$ , $\alpha = r^4 - r$ : $\dim Z_{1518}^2 = 2$ and contained in each target space, by an independent numeric implementation                                                                                                                                                                                                                                                                                        |
| Control                                                                                  | law 359, which 1518 implies universally, is contained with a certificate; law 3 (idempotence) is not satisfied by the linear magma over $R$ , as expected                                                                                                                                                                                                                                                                                                                                                                                  |
| Method check                                                                             | the same computation over the affine base $3x + y + 1 \pmod 6$ with fibre $\mathbb{Z}/2$ reproduces the 12-element $879 \not\equiv 4065$ model                                                                                                                                                                                                                                                                                                                                                                                             |
| Census for Corollary 2                                                                   | Mace4 2009-11A; sizes 2–4 complete labelled enumeration (audited against plain enumeration at sizes 2, 3), sizes 5–7 complete up to isomorphism (audited at size 4): one-generated classes at size 3: one (the shift); at sizes 4–7: none, out of 54, 632, 41 319, 8 393 390 models. Independently: z3 4.16, reachability encoding, sat only at size 3; labelled counts 1, 6, 54, 1500 at sizes 2–5 agree between the tools; z3 also gives unsat at size 8 (532 s). All enumerated 1518-magmas of size $\leq 7$ satisfy 47, 614, 817, 3862 |

| Layer (continued)    | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Census, Lean-checked | <code>census_lean/Census1518_n.lean</code> , $n = 2, \dots, 8$ : bitvector encoding of the table, the law at every pair and the reachability chain from an arbitrary generator; proved by <code>bv_decide</code> (bundled CaDiCaL, LRAT proof checked by Lean’s verified checker run natively; recorded as the per-theorem axiom <code>_native.bv_decide.ax</code> , the <code>native_decide</code> trust model, outside the lab’s allowlist). Size 3 is proved to be exactly the two labelled copies of the shift. Each file also carries encoding controls proved by kernel <code>decide</code> with no axioms: a genuine 1518 table of that size satisfies every encoded law hypothesis, and at size 3 the shift with generator 0 satisfies the reachability chain with <code>full = true</code> . The controls exist because a first size-8 file was vacuous ( $8\#3$ wraps to $0\#3$ , so its range hypotheses were false); the corrected size-8 file omits ranges, and its main theorem carries the SAT axiom like the others. Compile times 0.5 s to 68 s for sizes 2 to 7, 7:24.45 for size 8. <code>CensusBridge.lean</code> restates size 2 for functions $\text{Fin } 2 \rightarrow \text{Fin } 2 \rightarrow \text{Fin } 2$ and proves it by kernel <code>decide</code> with no native evaluation |

A Lean proof of Theorems 1 and 3 themselves would need the extension defined generically, the derivation of the rows from the law terms, and Step 1. None of that is done.

## 9 Scope and prior art

Not covered: infinite bases; coefficients  $\alpha_{x,y}, \beta_{x,y}$  varying with the base pair, which Section 5 shows do reach the targets over this very base; iterated extensions; fibres on which  $\beta$  is not injective.

No statement of Theorem 1 was found in the pinned ETP blueprint, paper, commentary or Lean sources, in the Lean Zulip *Equational* stream, in the SAIR Foundation Zulip, or on arXiv (searches dated 4 September 2026). What exists is the empirical record above. Originality is therefore unresolved, not claimed.

## 10 Questions

1. Is the vanishing  $H_{1518}^2(\mathbb{Z}/3 \text{ shift}, M) = 0$  for all finite 1518-fibres, or the resulting containment, already known or written down somewhere?
2. Theorem 4 proves the classification conjectured in 485148288. Was it proved anywhere before?
3. Over the shift the obstruction does *not* survive base-dependent coefficients: **Refutation939** is such an extension. Has the cohomology with base-dependent coefficients (abelian objects in the slice category, in Bolan’s phrasing) been computed for the shift, so that the 15-element models are the generic case of something? Section 7 answers this over the shift; is the family of Theorem 6 known?
4. Does a finite 1518-magma exist whose squaring map has order other than 1, 3 or 12?

## 11 Reproduction

Standard-library Python, Mace4 and Lean 4.33.1; inputs pinned by commit. Outputs as run are recorded alongside the scripts.

- `fetch_inputs.sh`: pinned ETP law list and refutation tables; Parsimagma’s uncovered-pairs file.
- `run_mace4.sh`, `monogenic_1518.py`: the census behind Corollary 2.
- `l2_symbolic.py`: the ring, the cocycle rows, the lattice containment, the numeric cross-check; run once for each base.
- `l2_symbolic.py`, `h2_vanishing.py`: the certificates for Theorems 1 and 3; `L2Cert.lean`, `H2Cert.lean`: their kernel checks.
- `rederive_rows.py`: independent noncommutative re-derivation of the rows.
- `z3_census.py`: the census with z3.
- `unreachable_set.py`: the 13 laws from the ETP finite graph.
- `witness939_varying.py`: the structure of Refutation939.
- `gen_census_lean.py`, `census_lean/`: the Lean census proofs and their README.
- `basedep_direct.py`, `basedep_orbits.py`, `basedep_family.py`: the enumeration, the orbit representatives and their exact lift (Section 7).
- `basedep_groebner2.py`, `basedep_groebner_modp.py`: the Gröbner bases over  $\mathbb{Q}$  and over  $\mathbb{F}_p$  (sympy 1.14).
- `family_vs_fifteen.py`, `family_more_fibres.py`, `family_squaring_morphism.py`, `gen_family_lean.py`: checks of the family and the Lean files.